Heinrich Heine Universität Düsseldorf

Report on the Research Internship: Investigating RAFT-Polymerization and PISA on Poly(Lipoates)

Faculty of Mathematics and Science
Heinrich Heine-University Düsseldorf

at the Chair of Macromolecular Chemistry, Düsseldorf

by:
Lena-Marie Aßmann
lena-marie.assmann@hhu.de
Student ID number: 3121504

as an Elective Module for the M. Sc. in Chemistry

Supervisor: Prof. Dr. G. Delaittre
Supervising Doctoral Student: S. Hylla, M. Sc.
Completion of the written report: 7th November 2026

Contents

1 Introduction **1**

1.1 Lipoic Acid . . . 1

1.1.1 General Background . . . 1

1.1.2 Esterification . . . 1

1.2 Reversible Addition Fragmentation Chain Transfer Polymerization . . . 2

1.2.1 General Background . . . 2

1.2.2 Application to Lipoates . . . 3

1.3 Polymerization Induced Self Assembly . . . 3

1.3.1 General Background . . . 3

1.3.2 Reverse Polymerization Induced Self Assembly . . . 3

1.3.3 Degradation of Poly(Lipoate) . . . 3

2 Experimental **4**

2.1 Methods . . . 4

2.1.1 Size-exclusion Chromatography . . . 4

2.1.2 Nuclear Magnetic Resonance . . . 4

2.1.2.1 Diffusion-ordered spectroscopy . . . 5

2.1.3 Matrix Assisted Laser Desorption/ Ionization Time Of Flight Mass Spectrometry . 6

2.1.4 Dynamic Light Scattering . . . 9

3 Results & Discussion **10**

3.1 Monomer Synthesis . . . 10

3.2 Thermal Reversible Addition Fragmentation Chain Transfer-Polymerization of Methyl Lipoate 10

3.3 Chain Extension & Polymerization Induced Self Assembly with Methyl Lipoate and *N,N*-dimethylacrylamide . . . 10

4 Summary & Outlook **11**

Acronyms **12**

List of Figures **13**

List of Tables **13**

Bibliography **15**

Appendix **A1**

AF4 . . . A1

1 Introduction

1.1 Lipoic Acid

1.1.1 General Background

Lipoic acid (LA) is a cyclic dithiol which is naturally occurring in both procaryotes and eukaryotes, and is displayed in Figure 1.1. LA has antioxidant properties and is a cofactor in the citric acid cycle. Its chirality provides two enantiomers wherein the (*R*)-(+)-LA is biologically active. The experimental data show a S-S bond length of 205.3 pm. It has lipophilic properties and can undergo ring-opening polymerization (ROP) which is why it has been selected as a monomer for this work.[1–3]

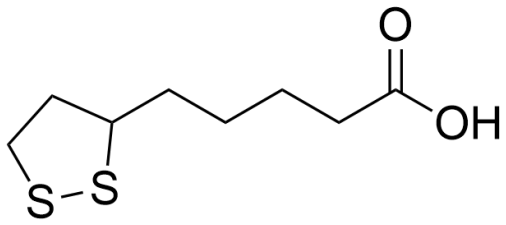


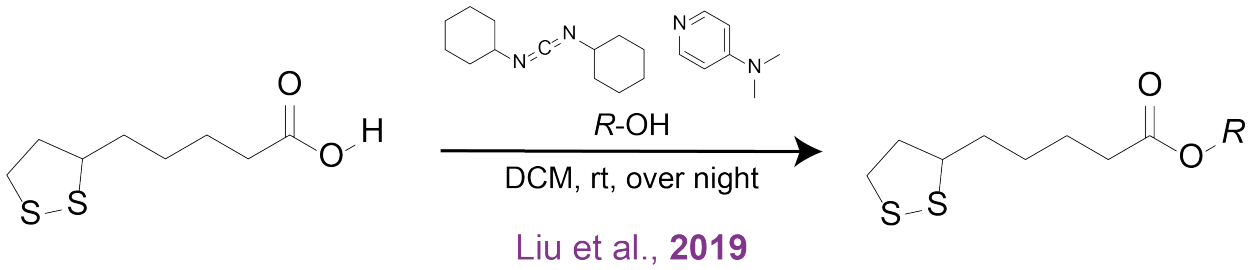
Figure 1.1: Chemical Structure of racemic lipoic acid (LA).

The ROP for LA can be anionic, cationic or radical which is why storing LA under UV protection and in the fridge is essential to prevent unwanted polymerization. Additionally, the carboxylic acid can be functionalized to obtain hyperbranched polymers.[4–8]

1.1.2 Esterification

To prevent ionization of the carboxylic acid and therefore unwanted polymerization, esterification is a method to modify the molecule in a minor way to retain most of its chemical properties for ROP, but without the carboxylic acid. With LA, the chosen mechanisms are either Fischer or Steglich esterification. The Fischer esterification is an acid catalyzed esterification under reflux. As a solvent, the alcohol is often used in excess. Here, the carbonyl group is activated by the catalyst and attacked by the nucleophilic alcohol. Water is eliminated after the alcohol addition which results in the ester.[9]
Alternatively, the Steglich esterification operates with a carbodiimide as a coupling agent at low temperatures between 0 °C to 20 °C and often 4-dimethylaminopyridine (DMAP) as an organocatalyst. Traditionally, dichloromethane (DCM), *N,N*-dimethylformamide (DMF), tetrahydrofuran (THF) or acetonitrile are used as a solvent.[10,11]
For LA, most esters for ROP are the methyl or ethyl lipoate. The Steglich esterification has been the more reliable choice or this monomer, but most recently the Fischer esterification has been adapted for LA for a more green synthesis.[5]

Steglich Esterification



Fischer Esterification

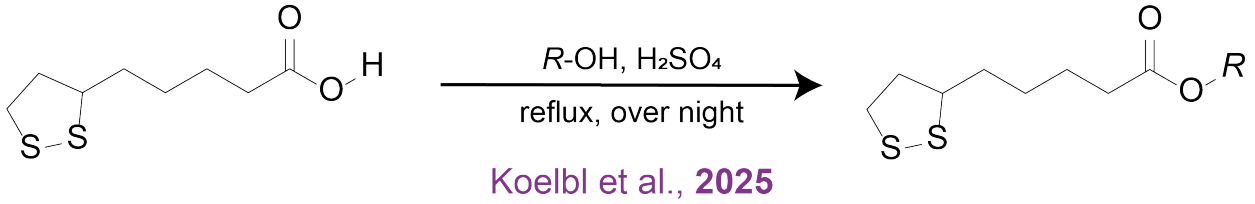


Figure 1.2: Overview of Steglich and Fischer esterification of lipoic acid (LA).

1.2 Reversible Addition Fragmentation Chain Transfer Polymerization

1.2.1 General Background

Reversible addition fragmentation chain transfer (RAFT) - equilibrium - "controlled" radical polymerization - pseudo-living polymerization - chain extension possible - initiation through chemical, thermal, photo initiation

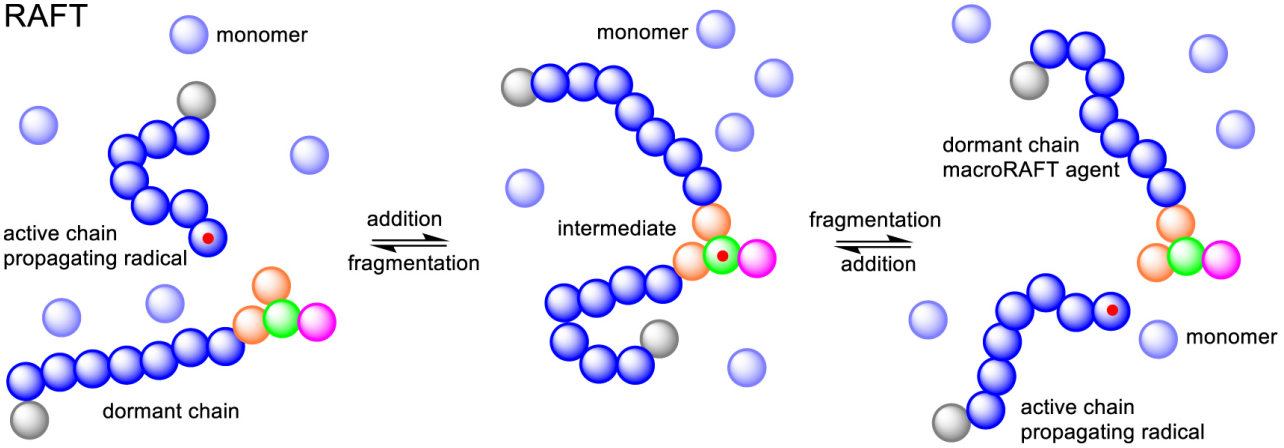


Figure 1.3: Visualization of addition and fragmentation of the RAFT mechanism.[12]

- description general structure

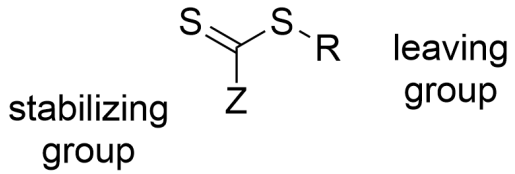


Figure 1.4: General structure of RAFT agents.

- established RAFT agents for commonly used monomers - compatibility issues for block copolymers

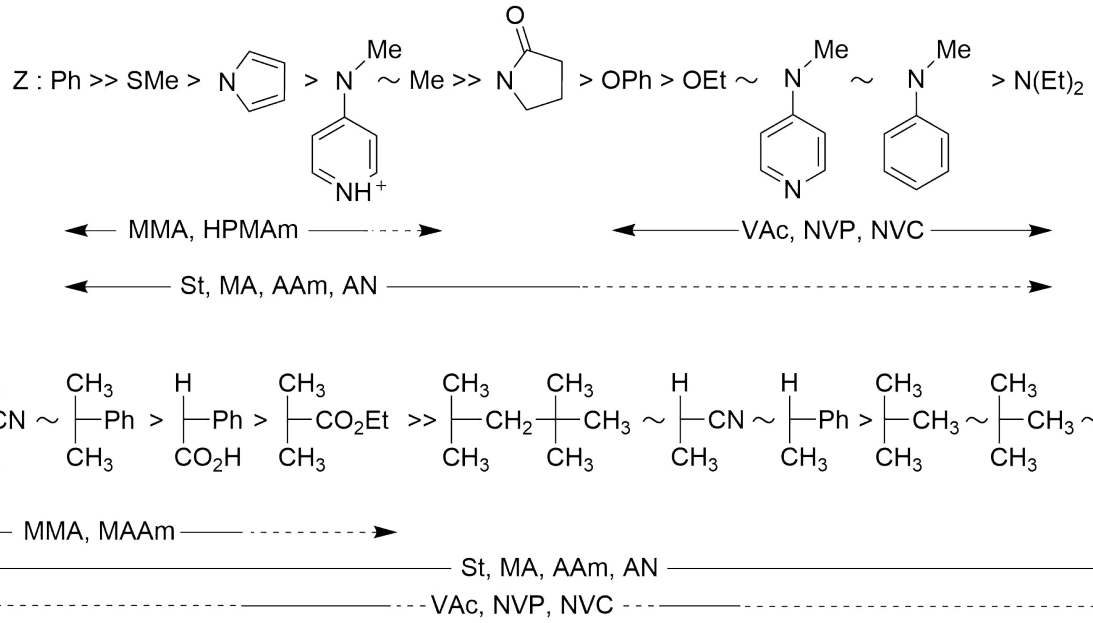


Figure 1.5: Different stabilizing and leaving groups for RAFT agents and their compatibility with common monomers.[13]

1.2.2 Application to Lipoates

- not common enough for established stabilizing and leaving groups - some found for lipoates[14,15]

1.3 Polymerization Induced Self Assembly

1.3.1 General Background

1.3.2 Reverse Polymerization Induced Self Assembly

1.3.3 Degradation of Poly(Lipoate)

2 Experimental

2.1 Methods

Within this work a variety of methods have been used. In chapter 1, some of the theoretical background for the reactions has been displayed, whereas the focus lies on the analytical aspects in this chapter. Different instruments have been used for monitoring the reaction course, as well as the chemical and physical properties of the monomers and polymers. The following sections give more insight on the selected methods and for some of them a description of the handling of the analysis data to obtain or display results. For writing this report, the translation tool *DeepL Translator* has been used for formulaic purposes.[16]

2.1.1 Size-exclusion Chromatography

Size-exclusion chromatography (SEC) is an analysis method to characterize polymers in regard to their molecular weight. This provides information on the molar mass distribution and characteristic values, such as the number average molar mass (M_n), mass average molar mass (M_w) and the dispersity ($Đ$).
The measurements for the SEC analysis were performed on the instruments within the Chair of Macromolecular Chemistry at Heinrich-Heine University (HHU) according to the established procedure within the group. The majority of the measurements have been collected on the instrument with DMF as a solvent, the remaining measurements have been made on the THF instrument. The data has been exported as an ASCII file and plotted on OriginPro 2025 v10.2.

2.1.2 Nuclear Magnetic Resonance

The technique nuclear magnetic resonance (NMR) spectroscopy can be used to determine the chemical composition, and by that also the purity, of chemicals, to obtain the conversion of a and monitor the polymerization reaction.
For the analysis, ^1H-NMR measurements were performed by the Center for Molecular and Structural Analytics @ Heinrich-Heine University (CeMSA@HHU) on the 600 MHz spectrometer. The results have been displayed and analyzed via the software MestReNova v17.0.1 which has been obtained through the chemistry department at Heinrich Heine University.

2.1.2.1 Diffusion-ordered spectroscopy

Diffusion-ordered spectroscopy (DOSY) is a form of NMR spectroscopy wherein the different signals are measured as a function of their chemical shift as well as their diffusion coefficient (D). This allows the differentiation between different molecules within a mixture which makes it a powerful analysis for analyzing block copolymers. Within the CeMSA@HHU, the measurments have been performed on the 600 MHz Evo Neo spectrometer. For the analysis, the software TopSpin 5.0.0 Processing by Bruker has been used. The software can be acquired through the academic processing license on the Bruker website.[17]
The analysis of DOSY spectra can be done the following way:

1. Make a copy of the folder with the raw data
2. Navigate to "*/pdata/1*" within the copied raw data folder and localize the file called "*2rr*"
3. Open TopSpin
4. Drag and drop the file "*2rr*" into TopSpin
5. Enter the following commands into the command line after the previous one is done:
    a) xf2
    b) abs2
    c) eddosy
    d) setdiffparm
    e) dosy2d
6. TopSpin can be closed once the commands have been run and MestReNova can be opened
7. The edited folder with the data can be dragged and dropped into MestReNova
8. In the *Stacked* tab, the button *DOSY Plot* needs to be pressed
9. The display of the spectra can be edited via the *Properties* window as usual

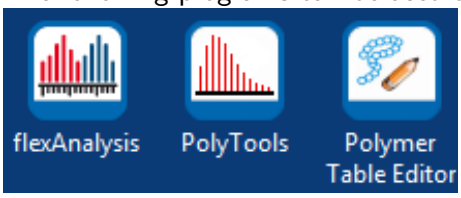
2.1.3 Matrix Assisted Laser Desorption/ Ionization Time Of Flight Mass Spectrometry

In order to receive information on the end groups of polymers, matrix assisted laser desorption/ ionization time of flight mass spectrometry (MALDI-ToF-MS) can be measured by the CeMSA@HHU. For this, the samples need to be in solution and an approximate molar mass is needed to set the measurement window. The concentration should be approximately $1\,\mathrm{mg\,mL^{-1}}$ and for the samples within this research work, THF has been selected as a solvent.
The analysis of the MALDI-ToF-MS spectra can be done as follows:

1. Request the measurement data from the CeMSA@HHU. This can be done via mail with specification of the measurement number which can be found on the handout result sheet

2. Copy the measurement data onto a USB stick and access the mass spectrometry computer within the Chair of Macromolecular Chemistry at HHU

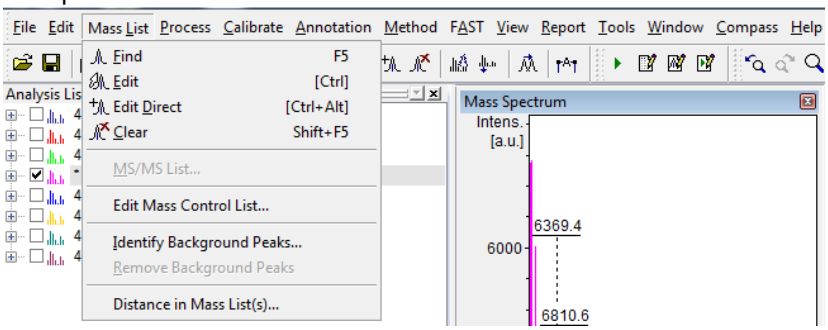
3. The following programs can be used to analyze MALDI-ToF-MS data of polymers:



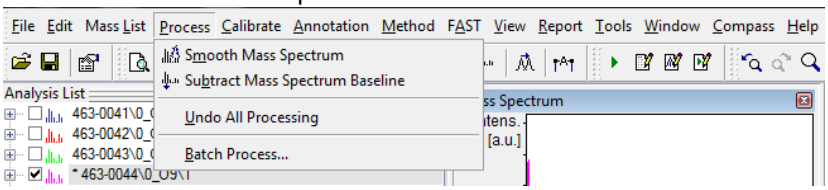
   a) flexAnalysis: Edit the peaklist

      i. Open your measurement data under *File – Open*

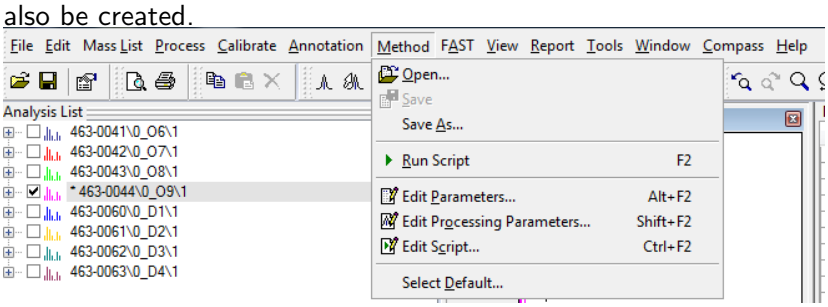
      ii. The peak list can be edited in the tab *Mass Liss*.



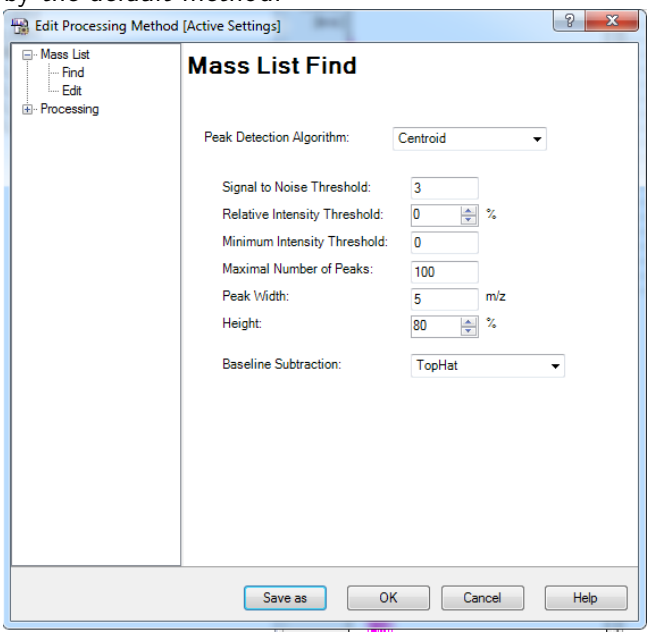
      iii. In the tab *Process* the spectra can be smoothed and a baseline can be substracted.



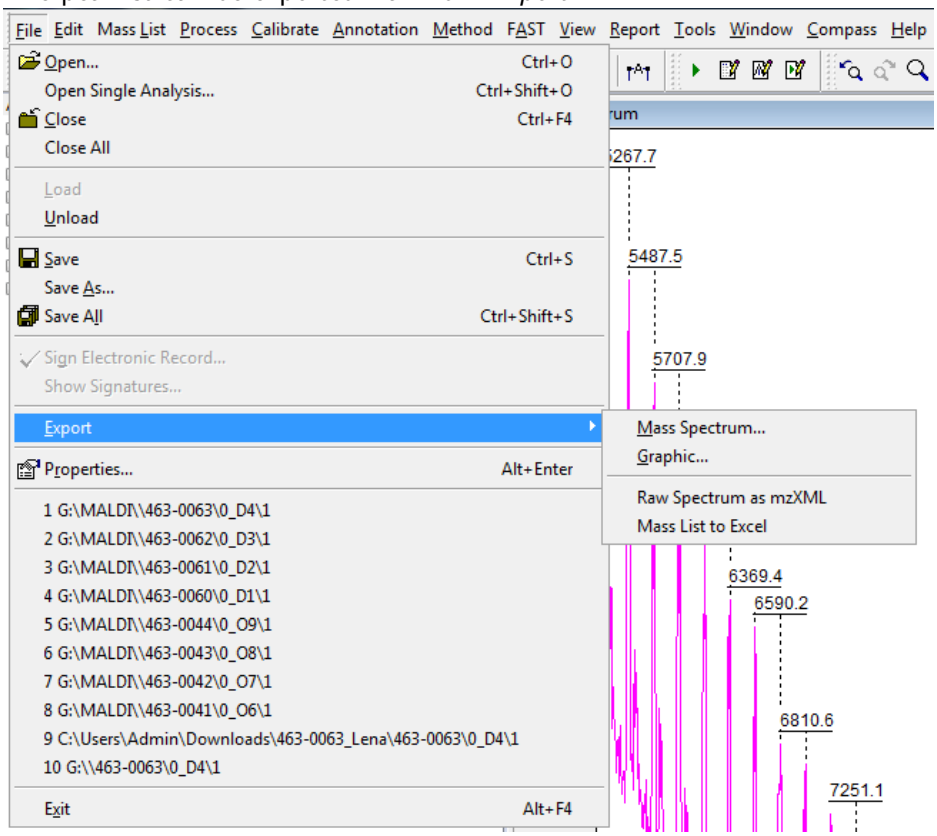
      iv. The parameters of the peak detection can be edited via *Method*. A script for analysis can also be created.



v. Within *Method – Edit Processing Method* different settings for detecting signals can be edited. This can be useful if many signals are in a narrow range and cannot be recognized by the default method.



vi. The peaklist can be exported via *File – Export*

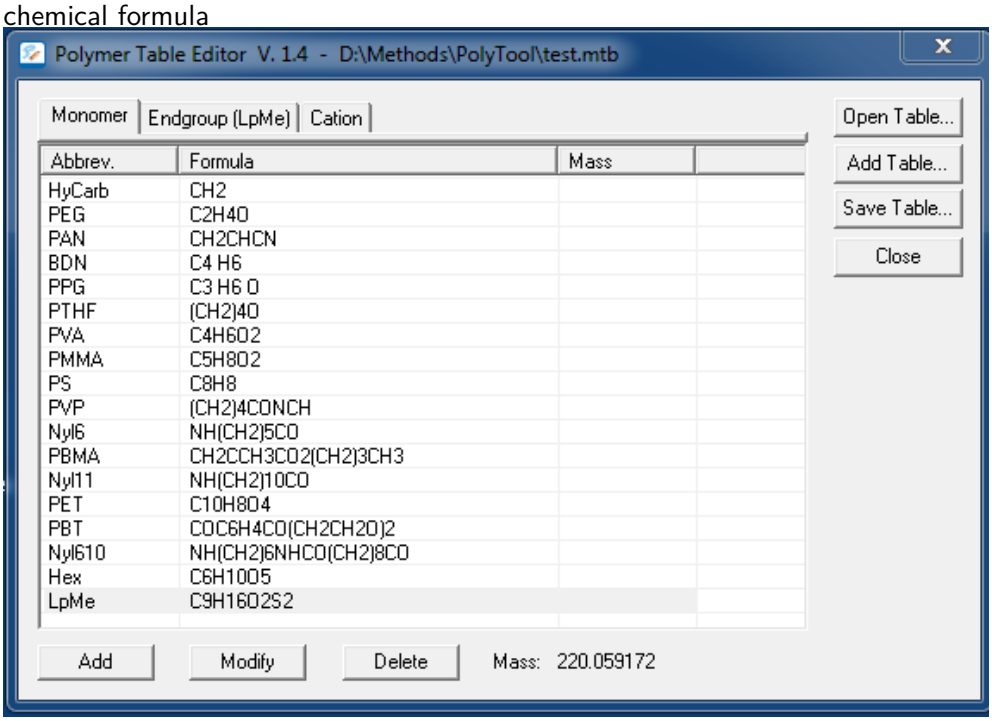


vii. Save your edits under *File – Save*

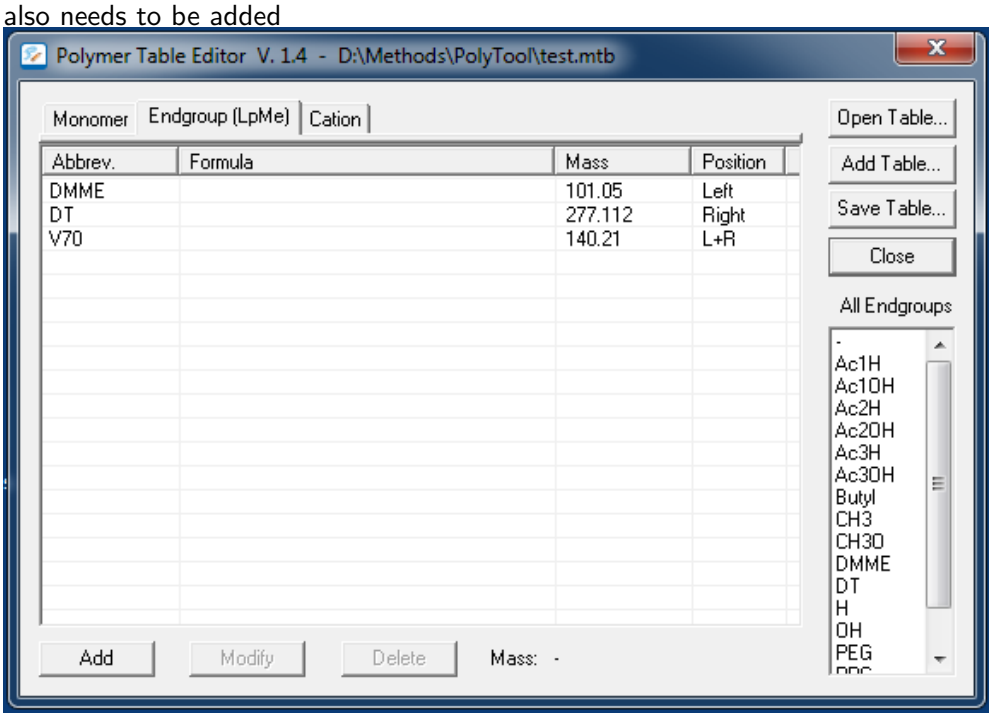
b) Polymer Table Editor: Create tables for your experiments with the masses for your specific monomers and end groups

i. The program has a list of standard monomers, endgroups and cations that can be edited

ii. In the tab *Monomer* new monomers can be added. By entering the chemical formula the mass is automatically calculated. Alternatively the mass can be entered without the chemical formula



iii. For each monomer, end groups can be defined by clicking on the respective row and the tab *Endgroup (abbrev.) – Add*. The position of the end group in relation to the monomer also needs to be added



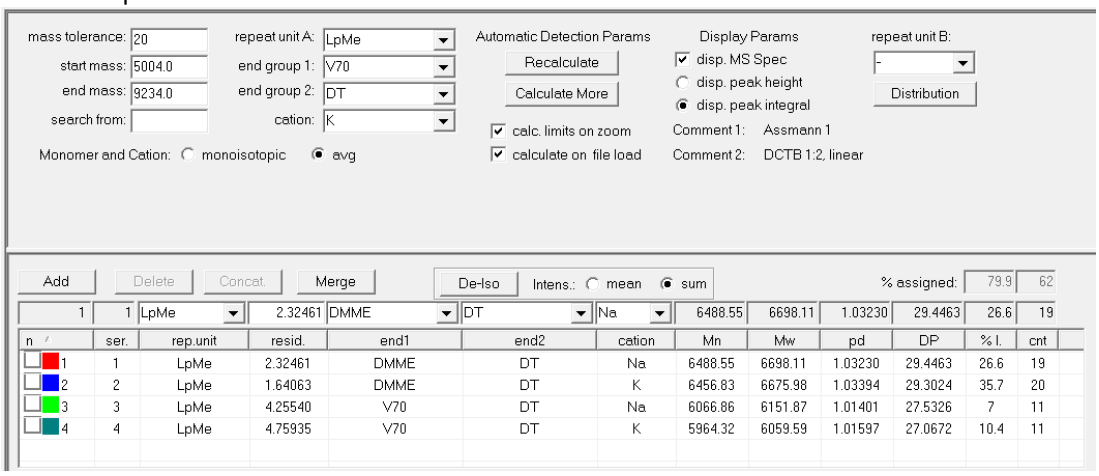
iv. Additional cations can be added or modified in the *Cation* tab if necessary

v. Click *Save Table...* and save it at an accessible location for later use

c) PolyTools: Analyse your polymer mass spectra by assigning the masses to peaks

i. Load the peaklist from *flexAnalysis* and the table from *Polymer Table Editor* into *PolyTools*

ii. The picture below shows the relevant software interface to analyze your polymer MALDI-ToF-MS spectra



A. The table at the bottom shows all the detected signals in which monomers, end groups and cations need to be assigned. The *residuals* are important to assign the correct signals. By checking the boxes in the first column, the entered monomers, end groups and cations become locked and changes cannot be made

B. Monomers, end groups and cations can be varied either at the top or immediately above the table

C. By clicking *Recalculate*, the *residuals* are getting calculated

D. By entering a *repeat unit B*, a second monomer for copolymers can be added

E. If the residuals are similar enough, signals can be *Merge*d

2.1.4 Dynamic Light Scattering

Dynamic light scattering (DLS) is a method for determining the hydrodynamic radius (r_h) of nano- and microparticles in solution. The measurements can be done on the instrument within the Chair of Physical Chemistry I and according to the instructions that can be provided by the Chair of Marcomolecular Chemistry at HHU. Adding to said instruction, it is important to export the data separated by a tab instead if a comma.

For this work, the raw data and the size distribution have been exported. The raw data entails some of the measurement parameters, such as the dynamic viscosity (η), absolute temperature (T), the number of runs and their duration. Additionally, the r_h of the species within the sample and the polydispersity index (PDI) can be found. The size distribution data is necessary for plotting the size distribution. For this, the data of a single run, which is displayed in one row, is copied into Microsoft Excel. The second half of that row entails the intensity data points. These are cut and pasted below the radii. The edited data can then be paste transposed and plotted in OriginPro 2025 (v10.2).

3 Results & Discussion

3.1 Monomer Synthesis

Fischer Steglich

3.2 Thermal Reversible Addition Fragmentation Chain Transfer-Polymerization of Methyl Lipoate

Different Conditions Different RAFT agents

3.3 Chain Extension & Polymerization Induced Self Assembly with Methyl Lipoate and *N,N*-dimethylacrylamide

4 Summary & Outlook

Acronyms

CeMSA@HHU	Center for Molecular and Structural Analytics @ Heinrich-Heine University
DCM	dichloromethane
D	diffusion coefficient
$Đ$	dispersity
DLS	dynamic light scattering
DMAP	4-dimethylaminopyridine
DMF	*N*,*N*-dimethylformamide
DOSY	diffusion-ordered spectroscopy
HHU	Heinrich-Heine University
LA	lipoic acid
MALDI-ToF-MS	matrix assisted laser desorption/ ionization time of flight mass spectrometry
M_n	number average molar mass
M_w	mass average molar mass
NMR	nuclear magnetic resonance
PDI	polydispersity index
RAFT	reversible addition fragmentation chain transfer
r_h	hydrodynamic radius
ROP	ring-opening polymerization
SEC	size-exclusion chromatography
T	absolute temperature
THF	tetrahydrofuran
η	dynamic viscosity

List of Figures

1.1 Chemical Structure of racemic lipoic acid (LA). 1
1.2 Overview of Steglich and Fischer esterification of lipoic acid (LA). 2
1.3 Visualization of addition and fragmentation of the RAFT mechanism.[12] 2
1.4 General structure of RAFT agents. 3
1.5 Different stabilizing and leaving groups for RAFT agents and their compatibility with common monomers.[13] 3

List of Tables

Bibliography

[1] A. R. Smith, S. V. Shenvi, M. Widlansky, J. H. Suh, T. M. Hagen, "Lipoic Acid as a Potential Therapy for Chronic Diseases Associated with Oxidative Stress", *Current Medicinal Chemistry* **2004**, *11*, 1135–1146, DOI 10.2174/0929867043365387.

[2] F. A. Moura, K. Q. d. Andrade, J. C. F. d. Santos, M. O. F. Goulart, "Lipoic Acid: Its Antioxidant and Anti-Inflammatory Role and Clinical Applications", *Current Topics in Medicinal Chemistry* **2015**, *15*, 458–483, DOI 10.2174/1568026615666150114161358.

[3] E. A. Rusu, K. Magyari, L. Baia, M. Baia, "Vibrational analysis of α-lipoic acid and its adsorption behavior study by SERS", *Journal of Molecular Structure* **2022**, *1248*, 131501, DOI 10.1016/j.molstruc.2021.131501.

[4] Y. Liu, Y. Jia, Q. Wu, J. S. Moore, "Architecture-Controlled Ring-Opening Polymerization for Dynamic Covalent Poly(disulfide)s", *Journal of the American Chemical Society* **2019**, *141*, 17075–17080, DOI 10.1021/jacs.9b08957.

[5] C. B. Koelbl, C. Obunadike, W. Ham, N. Mahmud, M. Garcia, E. Lizundia, J. C. Worch, "A Greener and More Scalable Synthesis of Biogenic Polydisulfides from Lipoic Acid", *ChemSusChem* **2025**, *18*, e202500194, DOI 10.1002/cssc.202500194.

[6] H. Tang, N. V. Tsarevsky, "Lipoates as building blocks of sulfur-containing branched macromolecules", *Polymer Chemistry* **2015**, *6*, 6936–6945, DOI 10.1039/c5py01005e.

[7] M. K. Debnath, N. Nupnar, Y. Yuan, E. R. Kone, V. O. Rodionov, M. J. A. Hore, M. Karayilan, "Dual-Degradable and Reprocessable Hyperbranched Polymers from Chain Transfer Agent-Lipoic Acid Conjugate", *Macromolecules* **2025**, *58*, 9930–9943, DOI 10.1021/acs.macromol.5c01498.

[8] K. R. Albanese, P. T. Morris, J. Read de Alaniz, C. M. Bates, C. J. Hawker, "Controlled-Radical Polymerization of α-Lipoic Acid: A General Route to Degradable Vinyl Copolymers", *Journal of the American Chemical Society* **2023**, *145*, 22728–22734, DOI 10.1021/jacs.3c08248.

[9] A. Mannu, A. Mele, "Fischer-Speier Esterification and Beyond: Recent Mechanicistic Advances", *Catalysts* **2024**, *14*, 931, DOI 10.3390/catal14120931.

[10] S. Munawar, A. F. Zahoor, S. M. Hussain, S. Ahmad, A. Mansha, B. Parveen, K. G. Ali, A. Irfan, "Steglich esterification: A versatile synthetic approach toward the synthesis of natural products, their analogues/derivatives", *Heliyon* **2024**, *10*, DOI 10.1016/j.heliyon.2023.e23416.

[11] A. Jordan, K. D. Whymark, J. Sydenham, H. F. Sneddon, "A solvent-reagent selection guide for Steglich-type esterification of carboxylic acids", *Green Chemistry* **2021**, *23*, 6405–6413, DOI 10.1039/d1gc02251b.

[12] C. L. Moad, G. Moad, "Fundamentals of reversible addition–fragmentation chain transfer (RAFT)", *Chemistry Teacher International* **2021**, *3*, 3–17, DOI 10.1515/cti-2020-0026.

[13] Let's learn RAFT polymerization remotely! | Wako Blog, Laboratory Chemicals | FUJIFILM Wako Pure Chemical Corporation, https://labchem-wako.fujifilm.com/us/wako-blog/030205.html (visited on 07/11/2026).

[14] D. Lee, H. Wang, X. Zuo, A. Alazmi, S.-Y. Jiang, R. Verduzco, "Light-Mediated Synthesis of Degradable Polymers With Complex Architectures Using α-Lipoic Acid as a Cleavable Comonomer", *Macromolecular Rapid Communications* **2026**, *47*, _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/marc.20260 e00006, DOI 10.1002/marc.202600006.

[15] K. Lee, S. Han, P. T. Morris, C. M. Bates, C. J. Hawker, C. Boyer, "Controlled Synthesis of Lipoate Homopolymers via Reversible Addition–Fragmentation Chain Transfer Polymerization", *Journal of the American Chemical Society* **2025**, *147*, 38796–38806, DOI 10.1021/jacs.5c14377.

[16] DeepL Translator, https://www.deepl.com/en/translator (visited on 26/10/2026).

[17] TopSpin | NMR Data Analysis, https://www.bruker.com/en/products-and-solutions/mr/nmr-software/topspin.html (visited on 26/10/2026).

Appendix

AF4